Saurabh Pawal
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B.Tech in CSE, 4th Year
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Education

- **National Institute of Technology, Warangal** *2022-2026*
  B.Tech in Computer Science Engineering

Experience

- **Web Development Intern**
  IMR Tech Solutions | Pune | Skills: Python, Django, LLAMA, PostgreSQL, PGvector May 2025 – July 2025
  - Built an **AI research assistant** using **Django, LangChain, LLaMA 2**, with **RAG pipeline** and vector search via **pgvector**.
  - Integrated scraping (**BeautifulSoup, newspaper3k**) and frontend in **Bootstrap 5** for file uploads and topic-based Q&A.
- **Software Development Intern**
  BLUESTOCK.in | Remote | Skills: C++, Python, Bootstrap, HTML, CSS May 2025 – July 2025
  - Programmed a full-stack IPO portal using **Django, DRF**, and **PostgreSQL** with APIs for listing data, PDFs, and live returns.
  - Responsive front end with **HTML, CSS, Bootstrap 5**; tested endpoints using **Postman** and managed code via **GitHub**.

Personal Projects

- **AI-Driven Failure Detection Simulation**
  Machine Learning-based Failure Prediction (Synthetic Data)
  - Built a failure prediction model using synthetically generated data to simulate equipment breakdown scenarios.
  - Implemented and evaluated **Logistic Regression and Random Forest** for binary classification.
  - Performed preprocessing and developed a local inference module to generate simulated failure alerts.
- **Learn Trade: Stock Trading Portfolio Management System**
  A simulation-based real-time management of stock market portfolio.
  - Engineered a **stock trading simulation tool** using **Python, Pandas, BeautifulSoup, and Tkinter**, featuring **real-time price scraping** from Google Finance.
  - Enabled **portfolio tracking, buy/sell operations**, CSV-based storage, and a user-friendly UI with live updates and account balance.
- **Melody Manager**
  Python-powered Music Library System with MySQL integration
  - **Crafted a full-featured music library system** with **MySQL** backend, supporting **CRUD operations** and advanced search by title, artist, album, and genre.
  - **Optimized query execution** for fast retrieval, enhancing performance and overall user experience.

Technical Skills and Interests

Languages: C++ , Python , HTML, CSS, SQL
Frameworks & Tools: Django , DRF , Bootstrap 5, ReactJS (Basic) , Github , Postman
Databases: PostgreSQL , MySQL , pgvector
Key Skills: REST APIs , Web Development , Fullstack Development , DSA , DBMS
AI Tech: LLaMA 2 , LangChain , BeautifulSoup, newspaper3k
Soft Skills: Leadership , Communication , Teamwork , Adaptability

Positions of Responsibility

- **Executive Lead**, Computer Science and Engineering Association *2022-Present*
  - Coordinated key initiatives focused on cutting-edge computer science technologies.
  - Led the design team in creating content for events, workshops, and promotions.
- **Secretary**, Team Avighna *2023-Present*
  - Supervised the collection of funds **(over 4 lakhs)** and played a vital role in all event transactions.
  - Coordinated with the team to manage footfall, accommodating 1500+ attendees at the event

Achievements

- cleared **National Talent Search Examination(NTSE) stage-2** with a score of **178/200**.
- Represented the state of Maharashtra as a **youth football player** for 5 years.
- Participated in **Oxford Presents Student's MUN** with junior college team as **sports captain**.